JAN–A–187

28 FEBRUARY 1945

JOINT ARMY-NAVY SPECIFICATION

ACID, PICRIC (TRINITROPHENOL)

Army Number	Navy Number
50-13-2B	51A24

This specification was approved by the War Department and the Navy Department for use of procurement services of the Army and the Navy and supersedes the following specification:

U. S. Army	Navy Department
50–13–2A	
8 Sept. 1938	----------

A. APPLICABLE SPECIFICATIONS.

A–1. The following specifications, of the issue in effect on the date of invitation for bids, form a part of this specification:

JOINT ARMY-NAVY SPECIFICATIONS

JAN–P–106—Packaging and Packing for Overseas Shipment, Boxes; Wood, Nailed.
JAN–P–122—Packaging and Packing for Overseas Shipment, Barrels, Slack

U. S. ARMY SPECIFICATIONS

50–0–1—General Specification for Ammunition except Small Arms Ammunition.[1]
100–2—Standard Specification for Marking Shipments by Contractors.

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[2]

FEDERAL SPECIFICATIONS

RR–S–366—Sieves, Standard, Testing.
QQ–S–781—Strapping, Flat, Steel.

INTERSTATE COMMERCE COMMISSION SPECIFICATION

2–L—Lining for Boxes.

B. GRADE.

B–1. This specification covers one grade of trinitrophenol known commercially as picric acid.

[1] Applicable only to Army purchases.
[2] Applicable only to Navy purchases.

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C. MATERIAL AND WORKMANSHIP.

C–1. See section E.

D. GENERAL REQUIREMENTS.

D–1. See section E.

E. DETAIL REQUIREMENTS.

E–1. *Color.*—White to yellow.

E–2. *Granulation.*—Minimum, 99.5 percent through 1410 micron (No. 14) U. S. Standard sieve.

E–3. *Solidification point.*—Minimum 120.0° C.

E–4. *Moisture.*—

E–4a. *When purchased dry.*—Maximum 0.20 percent.

E–4b. *When purchased wet.*—Maximum 22.0 percent.

E–5. *Ash.*—Maximum, 0.1 percent.

E–6. *Insoluble matter.*—Maximum, 0.1 percent.

E–7. *Acidity.*—

E–7a. *Total sulfuric acid.*—Maximum, 0.1 percent.

E–7b. *Nitric acid.*—None.

E–8. *Lead.*—Maximum, 0.0004 percent.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F–1. *Size of lots.*—A lot shall not exceed 100,000 pounds.

F–2. *Sampling.*—A minimum of 10 percent of the containers in the lot shall be selected in such a manner as to be truly representative of the lot. When lots comprise less than 100 containers, either 10 containers or all containers in the lot shall be selected. Approximately 1 pound shall be removed from each selected container by means of a thief or scoop. Each one of these primary samples shall be mixed thoroughly and a 4-ounce portion shall be placed in a bottle so labeled that the container from which the sample was taken can be easily identified. The remainder of these samples shall be mixed together thoroughly and quartered until a composite sample of approximately 1 pound is obtained. The composite sample shall be placed in a rubber-stoppered bottle and labeled to show the name of the material, manufacturer, plant, contract or order number, number of pounds in the lot, and lot number. All acceptance tests shall be made on the composite sample representative of the lot. The primary samples shall be held by the Government inspector for possible future examination should the composite sample fail to meet the requirements.

F–3. *Inspection.*—Inspection shall be made at the point of delivery unless otherwise specified in the contract or order.

F–4. *Tests.*—The following tests shall be made at a Government laboratory unless otherwise specified in the contract or order.

F–4a. *Granulation.*—Place a weighed portion of 50 gm. of the sample on a 1410 micron (No. 14) U. S. standard sieve to which a bottom pan has been attached. Place two metal washers on the sieve and shake for a maximum of 3 minutes or until no more material passes through the sieve. A standard vibrating machine may also be used for a period of 3 minutes using the tapper. Weigh any material remaining on the sieve and calculate the percentage passing through or retained on the sieve.

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F–4b. *Solidification point.*—Dry 80 to 100 gm. of the sample for 4 hours at 90° C. Transfer to the inner tube of the solidification point apparatus shown on figure 1 and melt. Place the tube in the apparatus and adjust the standardized centigrade thermometer so that the bulb is in the center of the liquid with a side thermometer in position for the emergent stem correction. Stir the molten material vigorously by hand or mechanically and carefully note the point where the temperature begins to rise when solidification begins. Record the temperature every 15 seconds until the maximum reading is obtained. Correct the maximum temperature for emergent stem by adding the value calculated from the formula:

$$N\ (T-t)\ 0.000159$$

where N=degrees in the exposed mercury column
T=the uncorrected solidification point
t=average temperature of the exposed mercury column determined by means of a second thermometer suspended so that its bulb is in the mid point of the exposed mercury column.
0.000159=the coefficient of expansion of mercury in glass.

Record the corrected reading to the nearest tenth of a degree as the solidification point of the sample.

F–4c. *Moisture.*—Transfer a weighed portion of approximately 5 gm. of the sample to a tared weighing dish of at least 1½ inches in diameter. Dry by heating in a vacuum oven at 65° C. for at least 4 hours or to constant weight. Cool in a desiccator and weigh. Calculate the loss in weight as percentage of moisture.

F–4d. *Ash.*—Transfer a weighed portion of approximately 5 gm. of the sample to a tared crucible. Moisten with sulfuric acid, and heat carefully until all carbonaceous residue has been eliminated. Add a few drops of nitric and sulfuric acids, heat to dryness, and ignite over a gas flame or in a muffle furnace. Cool the crucible and contents in a desiccator, and weigh. Calculate the increase in weight as percentage of ash.

F–4e. *Insoluble matter.*—Dissolve a weighed portion of 10 gm. of the sample in 150 ml. of boiling distilled water, continue the boiling for 10 minutes, filter while hot through a tared filtering crucible, wash thoroughly with hot water, and dry at 100° C. for 2 hours. Calculate the increase in weight as percentage of insoluble matter.

F–4f. *Acidity.*—

F–4f(1). *Sulfuric acid.*—Dissolve a known weight, approximately 10 gm. of the sample in 250 ml. of boiling distilled water. Acidulate with hydrochloric acid, and heat. Add a slight excess of a hot solution of barium chloride, with constant stirring. Allow to settle and filter while hot through a tared filtering crucible. Wash thoroughly with hot water, dry at 100° C. for 3 hours, cool in a desiccator, and weigh. Calculate the increase in weight to percentage of sulfuric acid in the sample.

$$\text{Percentage of sulfuric acid}=\frac{42.02A}{W}$$

where A=weight of barium sulfate precipitate
W=weight of sample.

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F-4f(2). *Nitric acid.*—Test a water solution of the sample with a solution of diphenylamine in concentrated sulfuric acid. Note if a blue coloration results, indicating the presence of free nitric acid.

F-4g. *Lead.*—Place a known weight, approximately 300 gm., of the sample in a 2-liter flask and allow to soak in 100 ml. of a hot saturated solution of barium hydroxide in 65 percent alcohol. Shake well, add 1,400 ml. of 95 percent alcohol, and digest at a temperature below the boiling point of alcohol until all the picric acid has been dissolved. Shake the solution thoroughly and allow to stand in the cold for several hours, or overnight, until most of the picric acid has crystallized out. Filter the solution without removal of the crystals of picric acid. Take 500 ml. of this filtered solution, which represents 100 gm. of picric acid, and add 10 ml. of a 0.10-percent solution of mercuric chloride in 95 percent alcohol. Add a few drops of nitric acid (just sufficient to render acid), and pass a slow stream of hydrogen sulfide through this solution for 15 minutes. Allow to settle 20 minutes, filter, and wash the precipitate with alcohol saturated with hydrogen sulfide. Dry the filter paper and ignite in a porcelain crucible at a dull red heat in order to drive off any salts of mercury present. Add 9 ml. of concentrated nitric acid and warm the mixture. Transfer the solution to a beaker by washing with sufficient water to make a total volume of 50 ml. Maintain the temperature at approximately 65° C. and electrolyze for 1 hour by means of a current of 0.4 ampere and 2.5 volts, using a tared platinum anode. Wash the anode by replacing the beaker, without interruption of the current, with one containing distilled water. Dry the anode at 100° C. for 2 hours, cool in a desiccator, and weigh. Calculate the percentage of lead in the sample.

$$\text{Percentage of lead} = \frac{86.62A}{W}$$

where A=weight of lead peroxide on anode
W=weight of sample.

F-5. *Retests.*—If the official sample representative of the lot fails to pass the inspection tests, the manufacturer shall have the option of having analysis of each primary sample made at his own expense. The manufacturer may then remove or replace defective portions of the lot represented by the primary samples which fail to meet the requirements, and submit the lot for acceptance, provided the markings on the container are such that complete removal or replacement of defective portions of the lot can be made to the satisfaction of the Government inspector.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT.

G-1. *Packing.*—

G-1a. *For domestic shipment.*—Unless otherwise specified, picric acid shall be packed in nailed wood boxes or barrels conforming to the requirements of Joint Army-Navy Specifications JAN-P-106 and JAN-P-122 respectively. The containers for dry picric acid shall be lined with a paper or bag conforming to the requirements of Interstate Commerce Commission Specification 2-L. Boxes shall be strapped in conformance with the strapping requirements of the appendix of Joint Army-Navy Specification JAN-P-106. The gross weight of containers shall not exceed approximately 100 pounds.

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G–1b. *For overseas shipment.*—Unless otherwise specified, picric acid shall be packed in nailed wood boxes or barrels conforming to the requirements of Joint Army-Navy Specifications JAN–P–106 and JAN–P–122 respectively. The containers for dry picric acid shall be lined with a paper or a bag conforming to the requirements of Interstate Commerce Commission Specification 2–L. Barrels shall be strapped and barrel heads shall be reinforced with suitable battens placed to receive ⅝-inch by 0.020-inch, nailless, flat steel strapping conforming to the requirements of Federal Specification QQ–S–781. Straps shall be stapled at intervals of approximately 6 inches, but the staples must not penetrate through the staves. Boxes shall be strapped in conformance with the strapping requirements of the appendix of Joint Army-Navy Specification JAN–P–106. The gross weight shall not exceed approximately 100 pounds.

G–2. *Marking.*—Unless otherwise specified, each container shall be plainly marked with the following information completed:

MATERIAL _____
SPECIFICATION NO. _____
STOCK NO. _____
MANUFACTURER _____
CONTRACT NO. _____
GROSS WEIGHT _____
DATE OF MANUFACTURE _____
LOT NO. _____

In addition, shipments for the Army shall be marked in accordance with the requirements of U. S. Army Specification 100–2; for the Navy, in accordance with the requirements of the latest issue of the Navy Shipment Marking Handbook.

H. NOTES.

H–1. Requests, requisitions, schedules, and contracts or orders should contain the title of the specification, the number and date.

H–2. *Uses.*—Picric acid covered by this specification is intended for use in the manufacture of ammonium picrate and of compositions used in the loading of high explosive shell.

H–3. Copies of Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C.

H–4. Copies of Joint Army-Navy specifications and Federal specifications (required for Army purchases) and U. S. Army specifications may be obtained, as indicated in the "Index of United States Army, Joint Army-Navy, and Federal Specifications used by the War Department." Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the War Department will obtain copies of Joint Army-Navy, United States Army, and Federal specifications through established War Department channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–5. Copies of Joint Army-Navy specifications and Federal specifications (required for Navy purchases) and Navy Department specifications may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command,

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Naval Supply Depot, Bayonne, N. J. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

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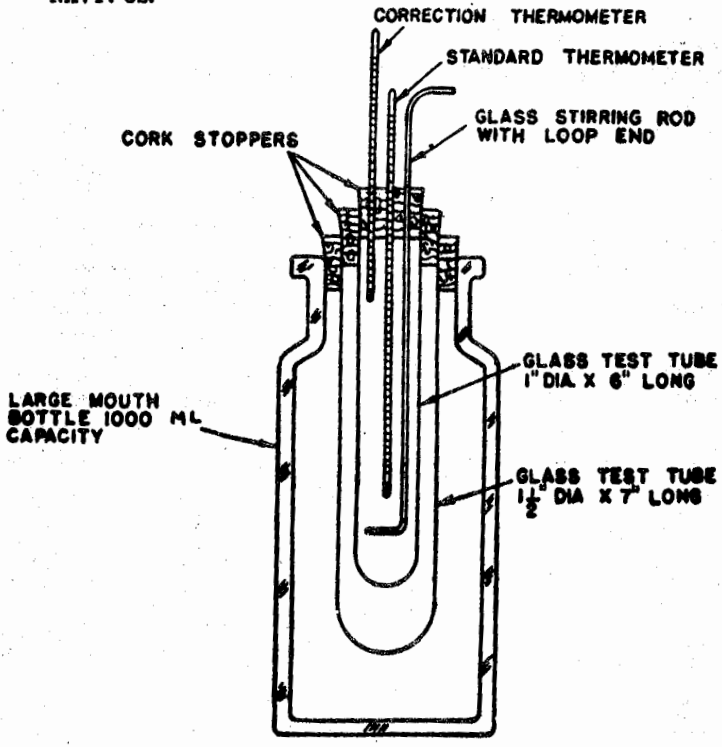


FIGURE 1.—Solidification-point apparatus.

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